Increased interoceptive accuracy and resistance to perceptual self-other illusion in frequent lucid dreamers support the predictive processing framework of lucid dreaming

Giacomina Grandolfo[1], Charline Urbain[1,2], Axel Cleeremans[1], Peter Simor[1,3]

[1] *Neuropsychology and Functional Imaging Research Group (UR2NF), Centre for Research in Cognition and Neurosciences (CRCN), UNI – ULB Neuroscience Institute, Université Libre de Bruxelles (ULB), Brussels, Belgium*

[2] *ULB, UNI, Laboratoire de Neuroanatomie et Neuroimagerie translationnelles (LN²T), Brussels, Belgium.*

[3] *Institute of Psychology, Eötvös Loránd University, Budapest, Hungary*

Abstract

Lucid dreaming (LD) enables conscious awareness during dreaming. Frequent lucid dreamers exhibit stable waking traits, such as greater frontopolar cortex volume and enhanced metacognition, suggesting that lucidity is associated with dispositional rather than only with sleep-specific factors. According to the predictive coding account of lucid dreaming, these trait-like capacities may reflect enhanced precision-weighting of self-related prediction errors, resolved through a superordinate self-model that stabilizes conscious experience during dreaming. We assumed that such mechanisms should also manifest during multisensory conflicts experienced in wakefulness.

We addressed this question using a cardio-visual enfacement illusion. Participants judged their resemblance to a set of self-other morphed faces before and after a stimulation where a picture of their face merged with an unfamiliar other's was overlaid with a pulsing shade, synchronously or asynchronously with their heartbeat. Two morphing levels (other-like and self-like) were used. Shifts were quantified via the point of subjective equality (PSE) before and after stimulation. Interoceptive accuracy (IAcc) was measured via heartbeat counting.

48 participants were tested (23 frequent lucid and 25 non-lucid dreamers); lucid dreamers showed higher IAcc. Synchronous stimulation reduced PSE values, replicating the enfacement illusion. This effect was modulated by IAcc: participants with lower interoceptive accuracy experienced the classic illusion with the other-like morph, while participants scoring with higher interoceptive accuracy experienced it with the self-like morph. Notably, lucid dreamers with low interoceptive accuracy showed the classic effect, but as interoceptive accuracy increased, they tended to reverse the illusion, indicating a resistance against the illusion. These findings offer the first behavioral evidence linking LD to self-consciousness, suggesting that interoceptive precision may shape and stabilize self-representations across states of consciousness.

Keywords: lucid dreaming, enfacement illusion, multisensory integration, interoception

Introduction (1119)

Lucid dreaming (LD) is a state of consciousness in which the sleeper becomes aware of the fact that the ongoing experience is self-generated while remaining immersed in the dream (van Eeden, 1913; LaBerge, 1980). This metacognitive insight distinguishes lucid from ordinary dreaming, where reflective monitoring is largely absent: dreamers rarely recognize their experience as internally produced, even when dream content deviates markedly from waking reality (Hobson et al., 2000; Nir & Tononi, 2010). While still poorly understood, the mechanisms enabling lucidity appear to rely partly on stable, trait-like capacities rather than sleep-specific processes. Frequent lucid dreamers show greater gray matter volume in frontopolar cortex (BA9/10), a region consistently linked to metacognitive monitoring, and exhibit stronger activation in this area during thought-monitoring tasks performed in wakefulness (Filevich et al., 2015). They also display increased functional connectivity between frontopolar and temporoparietal association cortices at rest (Baird et al., 2018). These neuroanatomical and neurofunctional differences suggest that lucid dreaming is not solely a sleep phenomenon but draws on cognitive resources that extend into waking cognition.

Predictive processing (PP) offers a framework for understanding how these capacities might operate. According to this model, the brain is a hierarchical system that continuously strives to predict the causes of its own sensory states (Friston, 2010; Clark, 2013). These predictions flow from higher levels of the system, which capture slow and abstract regularities, to lower levels, which encode the fast and detailed features of perception (Friston, 2010; Hohwy, 2013; Clark, 2016). Incoming sensory signals are evaluated against predictions, and any mismatches generate prediction errors that are propagated back up the hierarchy, prompting the system to revise its internal models (Friston, 2010; Clark, 2013). The balance between top-down predictions and bottom-up error signals is governed by precision weighting, which determines the informational impact of prediction errors at each level of the hierarchy (Hohwy, 2013; Clark, 2016).

This framework has been applied to dreaming as a working hypothesis: during REM sleep, sensory input is strongly attenuated and motor output suppressed by atonia, leaving the system to run mainly on internally generated predictions (Bucci & Grasso, 2017). Under these conditions, the system reduces the precision of lower-level sensory signals and relies primarily on mid- and high-level priors, which are more abstract and less connected to immediate perception (Bucci & Grasso, 2017). The predictive machinery continues to operate, but with minimal constraint from external sensory evidence, it "runs wild," exploring various combinations of its internal model (Windt,

2015). Windt (2015) describes dreaming as a form of mental improvisation: a spontaneous attempt to make sense of internally generated activity in the absence of reliable external feedback.

Simor et al. (2022) extended this account by proposing a multicomponent framework for lucid dreaming that foregrounds the role of interoceptive processing and multisensory integration. In their model, transient increases in arousal during sleep partially reinstate attentional resources, allowing interoceptive and proprioceptive signals to be processed with increased precision. This increased precision generates prediction errors that propagate forward through the processing hierarchy, reaching supramodal prefrontal and parietal regions otherwise relatively quiescent during REM sleep (Muzur et al., 2002; Maquet et al., 2005). Rather than producing the gradual model updating typical of non-lucid dreaming, these forwarded errors are resolved by the construction of a superordinate self-model: a metacognitive representation that integrates ambiguous signals from both sensory periphery and higher-order cortical regions into the coherent insight that the ongoing experience is a dream (Simor et al., 2022). In Bayesian terms, the system compares competing hypotheses, being awake versus dreaming, and selects the one that minimizes prediction error most effectively (Bucci & Grasso, 2017; Szymanek, 2021). Crucially, prediction errors alone may trigger the initial lucidity moment but appear insufficient to sustain it; the transient lucid state can easily shift back into non-lucid dreaming or end in abrupt awakening (LaBerge & Rheingold, 1991). Maintaining lucidity is proposed to depend on sustained multisensory integration, which involves combining bodily and environmental cues into a coherent self-representation (Simor et al., 2022). This process depends on both bottom-up stimulus features and top-down attentional control (Talsma, 2015), which, in the context of lucid dreaming, flexibly regulates the gain of prediction errors across processing levels, stabilizing the dream scenario while preserving the metacognitive self-model (Simor et al., 2022).

If frequent lucid dreamers possess a more precise self-model and more effective precision weighting of self-related prediction errors, these characteristics should be detectable as stable traits during waking tasks that engage the same underlying mechanisms. Specifically, two cognitive capacities should differentiate frequent lucid dreamers from non-lucid dreamers: (a) more precise interoceptive and multisensory processing, reflecting enhanced precision weighting of bodily prediction errors, and (b) greater resistance to perturbations of the self-model, reflecting a more stable self-referential representation. These capacities parallel what has been termed reality monitoring, the ability to distinguish internally generated from externally derived information (Simons et al., 2017). We extend this notion to real-time perceptual processing, predicting that these capacities should be observable through waking paradigms that place the self-model under controlled multisensory conflict.

We address this question using a cardio-visual version of the enfacement illusion, adapted from the paradigm introduced by Sel et al. (2017). In this task, participants judged their resemblance to a series of self-other morphed faces before and after a

stimulation phase in which a pulsing shade, synchronous or asynchronous with their heartbeat, was projected onto a face that merged their own features with those of an unfamiliar other. Two morphing levels were used for the stimulation (other-like and self-like), and shifts in self-face recognition were quantified by comparing the point of subjective equality (PSE) before and after stimulation. Interoceptive accuracy (IAcc) was measured via heartbeat counting. Sel et al. (2017) showed that synchronous cardio-visual stimulation increased self-identification with the morphed face, and that this effect scaled linearly with IAcc: participants with more precise cardiac perception displayed stronger blurring of self-other boundaries. This paradigm is ideally suited to our research question because it simultaneously engages interoceptive processing, multisensory integration, and self-other distinction, the three domains in which the PP account of lucid dreaming (Simor et al., 2022) predicts trait-level differences between frequent and non-frequent lucid dreamers.

Under the PP hypothesis, the mechanisms supporting lucidity should strengthen the precision and impact of self-related prediction errors, effectively reinforcing reality monitoring. This enhancement should attenuate the enfacement illusion by constraining the drift of the self-model under conditions of multisensory conflict. We therefore predicted that synchronous cardio-visual stimulation would elicit an enfacement illusion across all participants, replicating Sel et al. (2017), but that frequent lucid dreamers would show smaller shifts in perceived self-other boundaries than non-lucid dreamers. Such a pattern, diminished illusion susceptibility in lucid dreamers, would constitute the first waking behavioral evidence that the predictive mechanisms thought to support dream lucidity also shape how the self-model responds to multisensory challenges during wakefulness.

Methods

Participants

Forty-eight participants (41 females; mean age = 21.2 years, SD = 6.0) were recruited from the Université libre de Bruxelles and received course credits for their participation. The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the Faculty of Psychology and Educational Sciences at the Université libre de Bruxelles (protocol no. 2215/2024). Participants were assigned to one of two groups based on their lucid dreaming frequency (see *Lucid dreaming assessment*).

In the self-face recognition task, data from 4 individuals were excluded because their global shift in point of subjective equality (PSE) deviated by more than ±2.5 SD from the group mean (Sel et al., 2017). The final sample comprised 44 participants (37 females; mean age = 21.0 years, SD = 6.0), including 22 frequent lucid dreamers and 22 non-lucid dreamers.

The sample size is comparable to, or larger than, previous between-group studies of frequent lucid dreamers (e.g., Baird et al., 2018, n = 14 fLD; Filevich et al., 2015, n = 31 fLD) and to the original enfacement paradigm (Sel et al., 2017, N = 29).

Lucid dreaming assessment

Participants completed a two-component assessment of lucid dreaming. First, they reported the frequency of their lucid dreams over the past year on a scale ranging from *never* to *several times per week*. Following established criteria (Schredl & Erlacher, 2004; Dresler et al., 2012), individuals reporting at least one lucid dream per month were classified as frequent lucid dreamers (fLD, n = 23; 18 females; mean age = 22.6 years, SD = 8.0). Those reporting less frequent or no lucid dreams were classified as non-lucid dreamers (nLD, n = 25; 23 females; mean age = 20.0 years, SD = 2.8). Because lucid dreaming is a relatively rare phenomenon, recurrence is considered an important marker of trait-level lucidity, enhancing the construct validity of this grouping criterion (Dresler et al., 2012).

Second, participants completed the Lucidity and Consciousness in Dreams scale (LuCiD; Voss et al., 2013) to characterize the phenomenological profile of their lucid dreams. Consistent with its psychometric validation, five items (7, 17, 20, 26, and 28) loading on the REALISM and NEGATIVE EMOTION factors, which do not reliably discriminate between lucid and non-lucid dreams, were excluded from the global index (Voss et al., 2013). The retained subscales (INSIGHT, CONTROL, THOUGHT, MEMORY, DISSOCIATION, POSITIVE EMOTION) collectively capture the cognitive and metacognitive dimensions of lucidity. Frequent lucid dreamers scored significantly higher on the LuCiD scale (M = 2.53, SD = 0.65) than non-lucid dreamers (M = 1.83, SD = 0.52), $t(42.3) = 4.09$, $p < .001$, Hedges' $g = -1.17$, confirming the validity of the group assignment.

Stimuli

Each participant's neutral-expression face was photographed and morphed with two unfamiliar, gender- and ethnicity-matched models selected from the Chicago Face Database (CFD; Ma et al., 2015), using Fantamorph (Abrosoft, v5.4.8) and further edited with Canva (www.canva.com). This produced two sets of images.

For the self-recognition task (phases 1 and 3), a continuum of 51 morphs was created for each participant, ranging from 20% self / 80% other to 80% self / 20% other in 2% increments. All images were converted to grayscale and cropped to remove external features (hair, ears, neck), with a black frame applied around the facial contour to ensure that recognition judgments relied exclusively on internal facial morphology, following standard procedures for enfacement illusion studies (e.g., Tajadura-Jiménez et al., 2013).

For the cardio-visual stimulation phase (phase 2), colored, non-mirror-reversed photographs were used. Following Sel et al. (2017), two morphing levels were used: a 40%-self/60%-other morph (hereafter "other-like") and a 60%-self/40%-other morph

(hereafter "self-like"). To avoid carry-over effects from one condition to the other, each participant was presented with two unfamiliar gender-matched models, one in the synchronous condition and one in the asynchronous condition. Models associated with the synchronous and asynchronous conditions were randomly assigned across participants. This yielded four experimental conditions: synchronous other-like, synchronous self-like, asynchronous other-like, and asynchronous self-like.

Materials

Participants were seated at a desk in a quiet testing room. Cardiac activity was continuously monitored using a portable 3-lead ECG device (M5Stack ECG Module, AD8232; M5Stack, www.m5stack.com). ECG signals were collected via the AD8232 analog front-end chip interfaced with a microcontroller. A custom Python routine detected R-waves in real time and controlled stimulus delivery, ensuring that visual pulses were time-locked to the participant's heartbeat. Similar real-time ECG acquisition and synchronization solutions using the AD8232 have been reported previously (Kanani & Padole, 2018). Stimuli were presented on a laptop screen at a viewing distance of approximately 60 cm.

Procedure

The main experimental procedure was organized in four blocks (Sel et al., 2017), each comprising the following sequence: Pre-Stimulation Face Recognition Task (phase 1), Cardio-Visual Stimulation (phase 2), and Post-Stimulation Face Recognition Task (phase 3) (Figure 1).

Each recognition task began with a 5-second presentation of the participant's own photograph (100% self), followed by a 5-second presentation of the unfamiliar model (100% other). Participants were informed that morphed combinations of these two faces would appear during the task. They then rated each of the 51 morphed images on a visual analog scale (VAS, 0-100) anchored at "least similar to me" (0) and "mostly similar to me" (100) (Figure 1, left and right panels). PSEs were derived by fitting individual ratings to a logistic function and extracting the midpoint (50%) of the resulting psychometric curve. Changes in PSE between pre- and post-stimulation tasks (ΔPSE) indexed shifts in self-other representation.

Between the two recognition tasks, participants observed a 2-minute stimulation phase. A periodic pulse of reduced luminosity was applied to the morphed self-other face, one of the two morphing levels (other-like or self-like) associated with one of the two unfamiliar models, while background luminosity remained constant (Sel et al., 2017) (Figure 1, center panel). Each pulse lasted 100 ms and was delivered either synchronously or asynchronously with the participant's heartbeat. In the synchronous condition, pulses were triggered 200 ms after each R-wave, a delay corresponding to the peak of subjective heartbeat perception (Brener et al., 1993; Suzuki et al., 2013). In the asynchronous condition, pulses followed the participant's own cardiac rhythm with each inter-beat interval randomly accelerated or decelerated by ±10%. Faster and slower asynchronous rhythms were counterbalanced across participants.

The combination of two morphing levels (other-like, self-like) and two synchrony conditions resulted in four experimental blocks, counterbalanced in order across participants. After each block, participants completed a brief questionnaire (Sel et al., 2017) assessing their subjective experience. Three items probed changes in self-other face representation (e.g., "It seemed like the other's face began to resemble my own face"; Tajadura-Jimenez et al., 2012a), two items assessed the subjective experience of the pulsing rhythm (e.g., "It seemed as if I was tuned with the rhythm of the flashing"; cf. Aspell et al., 2013), and one item assessed self-reported distraction. All items were rated on a VAS (0-100) anchored at "disagree" and "agree" (see Table 1). Participants were not informed of the cardiac synchrony manipulation; they were told that the ECG recordings measured cardiac responses to face perception.

Following Sel et al. (2017), participants then performed a heartbeat counting task (Schandry, 1981) to measure interoceptive accuracy (IAcc), which was subsequently included as a covariate in the main analyses. They were instructed to silently count their own heartbeats, without taking their pulse, across nine intervals of 25, 35, and 45 s (three repetitions per duration). IAcc was computed for each interval as 1 - (|recorded - counted| / recorded) and averaged across trials.

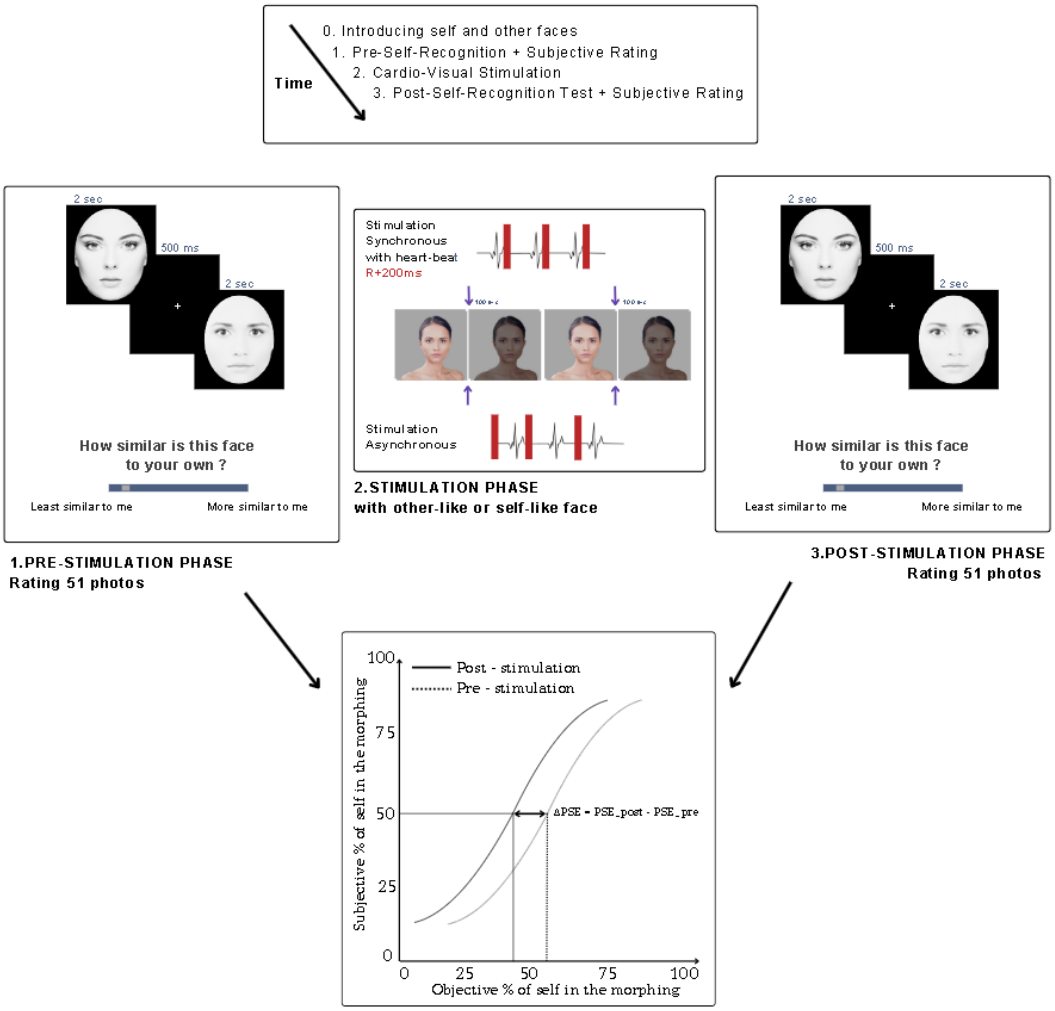

Figure 1. Schematic overview of the experimental procedure. Top: timeline of each block. Left panel: Pre-stimulation self-recognition task, in which participants rated 51 grayscale self-other morphed faces (20-80% self) on a visual analog scale. Center panel: Cardio-visual stimulation phase, in which a luminosity pulse (100 ms) was applied to a colored morphed face (other-like: 40% self, or self-like: 60% self) either synchronously (200 ms after R-wave) or asynchronously with the participant's heartbeat. Right panel: Post-stimulation self-recognition task, identical to the pre-stimulation task. Bottom panel: illustration of the ΔPSE metric. For each phase, similarity ratings were fitted with a logistic function; the point of subjective equality (PSE) corresponds to the midpoint (50%) of the fitted psychometric curve. ΔPSE = PSE_post − PSE_pre quantifies the stimulation-induced shift in self-other representation. Negative ΔPSE values indicate that morphed faces were perceived as more self-like after stimulation, reflecting the enfacement illusion.

Statistical analyses

All analyses were conducted in R v4.4.2 using the lme4, lmerTest, emmeans, afex, effectsize, and BayesFactor packages. For each participant and condition, similarity ratings were fitted with a logistic function (Sel et al., 2017) using nonlinear least-squares estimation, and the point of subjective equality (PSE) was extracted as the midpoint of the curve. PSE change scores were computed as ΔPSE = PSE post-stimulation – PSE pre-stimulation. Participants whose mean ΔPSE deviated by more than 2.5 SD from the group mean were excluded (Sel et al., 2017), yielding a final sample of 44 participants. Mean interoceptive accuracy was grand-mean centered to preserve between-group variance of theoretical interest (Enders & Tofighi, 2007). Sum-to-zero contrasts were applied for all categorical predictors, and type III tests were used for all ANOVAs and mixed models.
The manipulation check was assessed with a repeated-measures ANOVA aligning with the statistical procedure used in the original study (Sel et al., 2017) including Time, Synchrony, and Morphing as within-subject factors. To test whether interoceptive accuracy modulated PSE changes, we used a linear mixed-effects model rather than the ANCOVA of Sel et al. (2017), as the LMM framework properly accounts for the within-subject dependency across the four observations per participant while including interoceptive accuracy as a between-subject continuous covariate. The model included Synchrony and Morphing as fixed factors, mean-centered interoceptive accuracy as a continuous covariate, and participant as a random intercept. For direct comparability with Sel et al. (2017), we also replicated their original RM-ANCOVA.

To examine whether lucidity further moderated these effects, Lucidity was added as a between-subject factor in an additional model, with participant as a random intercept. A random-slope specification allowing Synchrony and Morphing effects to vary by participant was also tested: the model converged with a marginal warning. The core three-way Synchrony × Morphing × IAcc interaction remained highly significant ($p < .001$), while the exploratory Synchrony × IAcc × Lucidity interaction was attenuated ($p = .10$). Following the parsimony principle (Bates et al., 2015), the random-intercept model was retained for the main analysis (see Supplementary Materials).

Effect sizes are reported as partial eta-squared for ANOVAs, Hedges' g for pairwise comparisons, and Bayes factors (BF10) for the primary effects of interest. Analyses testing the replication of the enfacement illusion and its modulation by interoceptive accuracy were treated as confirmatory, whereas analyses involving lucid dreaming group status were treated as exploratory. Results were robust to group-mean centering of interoceptive accuracy (see Supplementary Materials). Data and analysis scripts are available at [OSF link].

Results

Manipulation check

PSE values were analyzed using a repeated-measures ANOVA with Time (pre vs. post), Synchrony (synchronous vs. asynchronous), and Morphing (other-like vs. self-like) as within-subject factors. A significant main effect of Time was observed, $F(1, 43) = 17.87$, $p < .001$, $\eta^2p = .294$, BF10 = 36.25, indicating a global reduction in PSE after stimulation: participants perceived morphed faces as more self-like following the cardio-visual phase. A significant Time × Morphing interaction, $F(1, 43) = 5.82$, $p = .020$, $\eta^2p = .119$, reflected a stronger PSE reduction for other-like morphs than for self-like morphs, indicating that the shift in perceived self-other similarity driven by the enfacement illusion was more pronounced when the morphed picture used for the stimulation contained a greater proportion of the other's features, consistent with Sel et al. (2017). A marginal main effect of Synchrony was observed, $F(1, 43) = 2.99$, $p = .091$, $\eta^2p = .065$, pointing to a trend toward stronger enfacement shifts under synchronous than asynchronous stimulation that reached significance only when moderated by interoceptive accuracy (see below), in line with Sel et al. (2017). No other effects reached significance (all $p > .16$). This pattern replicates the main findings of Sel et al. (2017), including the absence of a three-way Time × Synchrony × Morphing interaction, $F(1, 43) = 0.15$, $p = .696$.

Enfacement illusion: modulation by interoceptive accuracy

ΔPSE values were analyzed using a linear mixed model (LMM) with Synchrony and Morphing as within-subject fixed factors, mean-centered IAcc as a continuous covariate, and participant as a random intercept. This model revealed a significant main effect of Morphing, $F(1, 126) = 4.88$, $p = .029$, $\eta^2p = .037$, and crucially, a significant three-way Synchrony × Morphing × IAcc interaction, $F(1, 126) = 8.56$, $p = .004$, $\eta^2p = .064$, BF10 = 5.78, **replicating Sel et al. (2017) in showing that the influence of cardio-visual synchrony on perceived self-other boundaries depended on participants' interoceptive accuracy** (Figure 2).

To characterize this three-way interaction, we first examined how ΔPSE varied with IAcc in each cell of the Synchrony × Morphing design. Simple slope analyses revealed that the slope of ΔPSE as a function of IAcc was significantly negative in the synchronous self-like condition ($b = -19.25 \pm 6.23$, $p = .002$, BF10 = 8.49), indicating that higher interoceptive accuracy was associated with a stronger enfacement effect in this condition. No reliable modulation emerged in the other three conditions (all $p > .24$) (Figure 2). This pattern represents a partial divergence from Sel et al. (2017), who reported the strongest IAcc modulation on the synchronous other-like condition rather than on the self-like condition observed here.

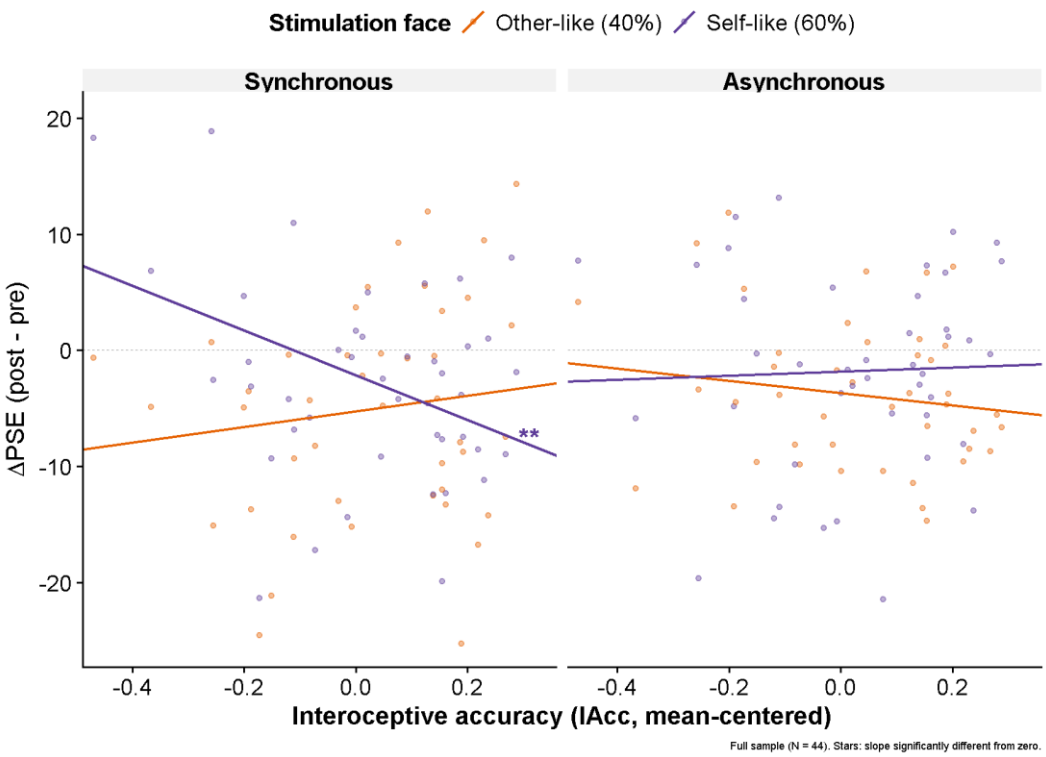

Figure 2. Modulation of the enfacement effect by interoceptive accuracy. Individual ΔPSE values (dots) and model-estimated slopes as a function of mean-centered interoceptive accuracy (IAcc), separately for cardio-synchronous (left) and cardio-asynchronous (right) stimulation, and for each morphing level (other-like: 40% self, orange; self-like: 60% self, purple). The slope of ΔPSE on IAcc was significantly negative in the cardio-synchronous self-like condition only (b = −19.25 ± 6.23, p = .002, BF10 = 8.49), indicating that the enfacement effect in this specific condition increased with interoceptive accuracy. **p < .01. Full sample (N = 44).

To further test whether the enfacement illusion was actually present at specific IAcc levels, we extracted estimated marginal means (EMMs) at low (−1.5 SD) and high (+1.5 SD) IAcc. At low IAcc, synchronous stimulation produced a significant ΔPSE in the other-like condition (−7.19 ± 2.25, p = .002, BF10 = 11.69) but not in the self-like condition (3.42 ± 2.25, p = .130), while asynchronous stimulation showed no significant effects (other-like: −2.16, p = .34; self-like: −2.33, p = .30). This replicates the classic enfacement pattern reported by Sel et al. (2017): at low IAcc, only synchronous stimulation with the other-like face shifted self-other boundaries. Consistent with this, the self-like morph, which Sel et al. treated as a control condition, showed no significant effect in low-IAcc participants.

At high IAcc, however, the pattern diverged from the classic enfacement effect: a significant ΔPSE emerged in the synchronous self-like condition (−7.52 ± 2.08, p < .001, BF10 = 43.62) and in the asynchronous other-like condition (−5.17 ± 2.08, p = .014, BF10 = 1.68), while the synchronous other-like condition, the only effective condition at low

IAcc, was not significant (−3.37 ± 2.08, p = .107) (Figure 3). Thus, while low-IAcc participants replicated the classic enfacement pattern of Sel et al. (2017), the high-IAcc pattern diverged: the illusion emerged on two distinct conditions (synchronous self-like and asynchronous other-like), neither of which was the canonical synchronous other-like effect. This heterogeneity at the upper end of the IAcc distribution raised the possibility of distinct response profiles within the sample, which we examine below by considering frequent lucid dreaming status as a grouping factor.

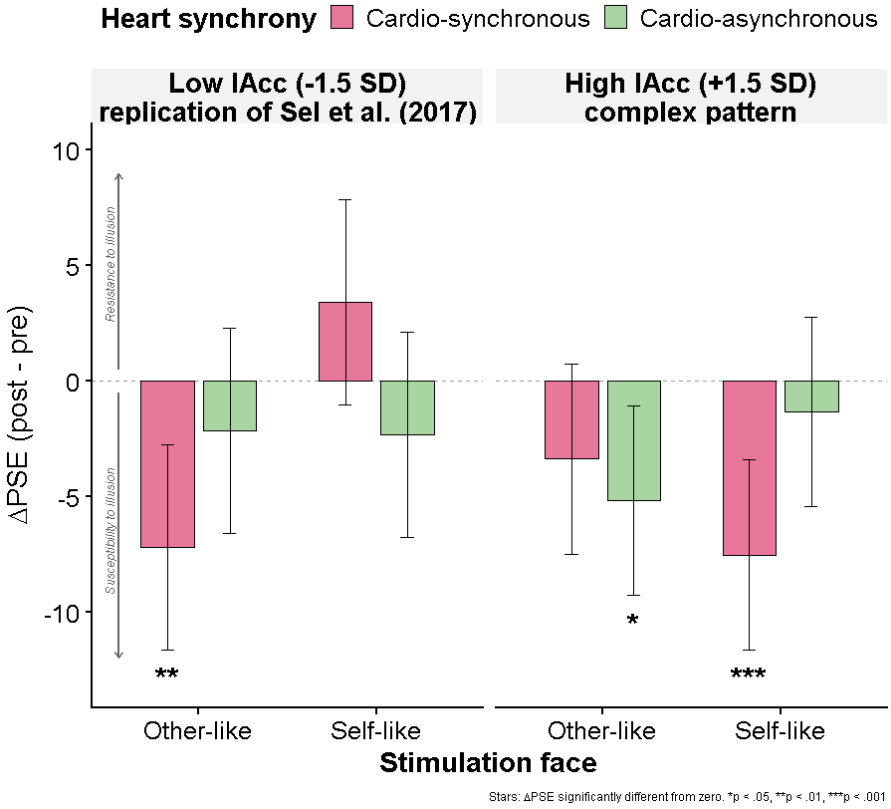

Figure 3. Enfacement illusion modulated by interoceptive accuracy. Estimated marginal means of ΔPSE (post minus pre) at low (-1.5 SD, left) and high (+1.5 SD, right) interoceptive accuracy (IAcc). Arrows on the y-axis indicate the direction of the effect: negative ΔPSE values reflect susceptibility to the illusion (morphed faces perceived as more self-like after stimulation), whereas positive values reflect resistance. At low IAcc, only cardio-synchronous stimulation with the other-like face produced a significant shift, replicating Sel et al. (2017). At high IAcc, the pattern diverged: both the cardio-synchronous self-like and the cardio-asynchronous other-like conditions produced significant shifts. Error bars represent 95% confidence intervals. *p < .05, **p < .01, ***p < .001.

Differential profiles in lucid and non-lucid dreamers

Because the replicated paradigm included the heartbeat counting task (Schandry, 1981), we had the opportunity to compare interoceptive accuracy between frequent and non-lucid dreamers for the first time, to our knowledge. Frequent lucid dreamers exhibited significantly higher IAcc than non-lucid dreamers (M_fLD = 0.72, SD = 0.16; M_nLD = 0.59, SD = 0.22; Welch's t(44.1) = 2.44, p = .019, Hedges' g = −0.69, BF10 = 2.87; Figure 4). Note that interoceptive accuracy (N = 48) was measured independently of the self-recognition task (N = 44) on which the exclusion criterion was based. Notably, the mean IAcc of our non-lucid subgroup (0.59) closely matched the overall mean reported by Sel et al. (2017) (0.60), whereas the inclusion of frequent lucid dreamers shifted the sample mean upward (0.66). Moreover, the balance of our sample (23/48 frequent lucid dreamers) constituted an intentional overrepresentation relative to the general population, where approximately 25% of individuals report at least one lucid dream per month (Schredl & Erlacher, 2004). This altered the IAcc distribution of our sample, suggesting that the IAcc-dependent pattern reported above might be explained by the overrepresentation in our sample of a profile whose cognitive traits, namely interoceptive precision and multisensory integration, are particularly relevant for the mechanisms underlying the enfacement illusion. To test this, Lucidity (fLD vs. nLD) was added as a between-subject factor to the LMM.

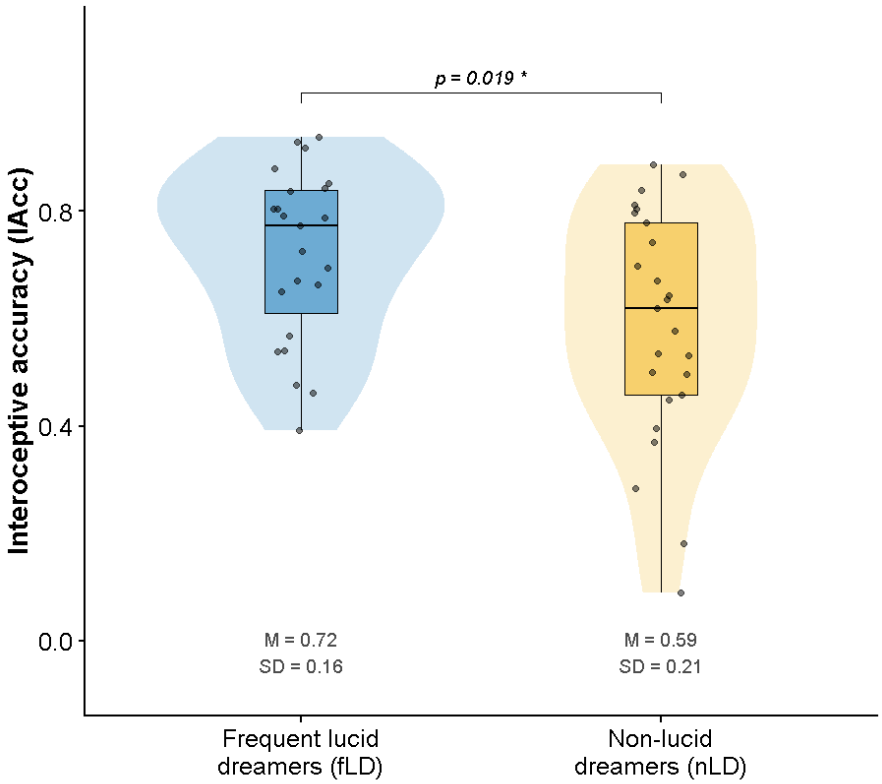

Figure 4. Interoceptive accuracy (IAcc) in frequent lucid dreamers (fLD, n = 23) and non-lucid dreamers (nLD, n = 25). IAcc was measured using the heartbeat counting task (Schandry, 1981) on the full sample (N = 48), as the exclusion criterion was based on PSE values only. Violin plots show the distribution of individual scores, boxplots indicate the median and interquartile range, and dots represent individual participants. Frequent

lucid dreamers exhibited significantly higher IAcc than non-lucid dreamers (Welch's t(44.1) = 2.44, p = .019, Hedges' g = 0.69, BF10 = 2.87).

This model revealed a significant main effect of Lucidity, $F(1, 40) = 5.86$, $p = .020$, $\eta^2p = .128$, indicating that lucid and non-lucid dreamers showed distinct overall ΔPSE profiles. The Synchrony × Morphing × IAcc interaction persisted, $F(1, 120) = 7.95$, $p = .006$, $\eta^2p = .062$, and a Synchrony × IAcc × Lucidity interaction also emerged, $F(1, 120) = 4.23$, $p = .042$, $\eta^2p = .034$, BF_inclusion = 3.3, suggesting that the two groups featured the synchrony-IAcc relationship differently. Decomposition of the simple slopes (figure 5) revealed that this interaction was carried exclusively by the synchronous conditions: no significant IAcc modulation emerged in the asynchronous conditions for either group (all $p > .19$). For clarity, Figure 5 therefore illustrates the synchronous conditions only, retaining the same grammar as Figure 2 but decomposed by lucid dreaming status.

Among frequent lucid dreamers ($n = 22$), ΔPSE showed a significant positive slope with IAcc in the synchronous other-like condition ($b = 27.88 \pm 10.61$, $p = .010$, BF10 = 2.41). This positive slope indicates more than a mere attenuation of the illusion: as interoceptive accuracy increased, frequent lucid dreamers actively inverted the enfacement effect, rating the morphed faces as less self-like than before stimulation, in the opposite direction to both the classic enfacement effect reported by Sel et al. (2017) and the pattern observed in our non-lucid dreamers. This active opposition to self-other blurring may, in wakefulness, echo the defining feature of lucidity itself, the capacity to reframe rather than passively accept an ongoing perceptual experience. No other slopes were significant in this group (all $p > .30$).

EMM decomposition confirmed this pattern: at low IAcc, lucid dreamers showed a significant enfacement effect in the synchronous other-like condition (-12.96 ± 4.18, $p = .002$, BF10 = 8.84), but at high IAcc, ΔPSE became positive and non-significant (2.89 ± 2.78, $p = .30$, BF10 = 0.14).

Among non-lucid dreamers ($n = 22$), a significant negative slope emerged in the synchronous self-like condition ($b = -26.57 \pm 7.76$, $p < .001$, BF10 = 23.62), amplifying the same IAcc-driven shift toward self-like sensitivity observed in the full-sample analysis. EMM decomposition showed that at low IAcc, non-lucid dreamers exhibited only a marginal trend toward enfacement in the synchronous other-like condition (-5.08 ± 2.59, $p = .052$), echoing the classic Sel et al. (2017) baseline, with no significant effect in the other three conditions (all $p > .16$). At high IAcc, however, all conditions except asynchronous self-like elicited a significant ΔPSE (synchronous other-like: -10.45, $p < .001$, BF10 = 23.77; synchronous self-like: -10.59, $p < .001$, BF10 = 27.36; asynchronous other-like: -8.07, $p = .009$, BF10 = 2.51), suggesting broadly blurred self-other boundaries in non-lucid participants with high interoceptive accuracy.

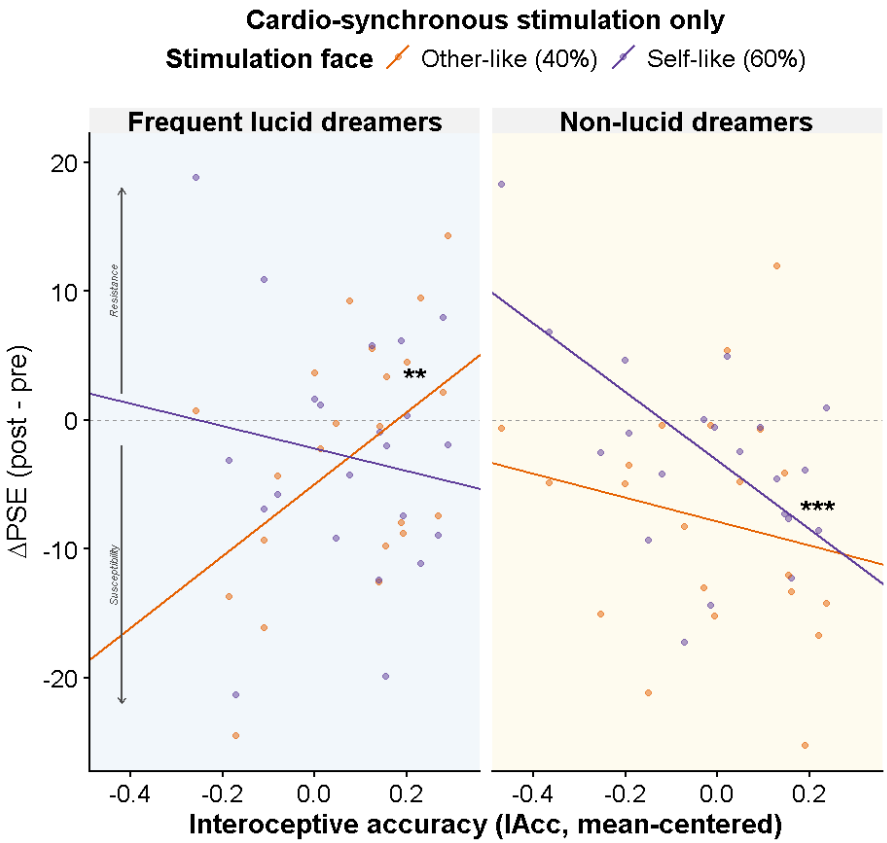

Figure 5. *Relationship between interoceptive accuracy (IAcc) and ΔPSE under cardio-synchronous stimulation, separately for frequent lucid dreamers (fLD, left, blue background) and non-lucid dreamers (nLD, right, yellow background). Lines represent model-estimated slopes from the linear mixed model including Lucidity, separately for each stimulation face (other-like: 40% self, orange; self-like: 60% self, purple). Individual dots represent observed ΔPSE values. Arrows on the y-axis indicate the direction of the effect: positive ΔPSE values reflect resistance to the illusion, whereas negative values reflect susceptibility. Among fLD, the other-like condition showed a significant positive slope (p = .010), indicating resistance to the enfacement illusion as IAcc increased. Among nLD, the self-like condition showed a significant negative slope (p < .001), indicating stronger illusion at higher IAcc. **p < .01, ***p < .001.*

Subjective reports

After each block, participants rated three enfacement items (Q1: "It felt like the other's face was my face"; Q2: "It seemed like the other's face began to resemble my own face"; Q3: "It seemed like my own face began to resemble the other person's face"), two pulsing items (Q4-Q5), and one distraction item (Q6) on a VAS (0-100).

Following the analysis strategy of Sel et al. (2017), an ANOVA with 3 Questions (Q1; Q2; Q3) × 2 Synchrony (synchronous; asynchronous) × 2 Morphing (other-like; self-like) was carried out. Results revealed a main effect of Question, $F(1.78, 76.59) = 10.57$, $p < .001$, and a Question × Synchrony interaction, $F(1.64, 70.70) = 4.37$, $p = .022$; a pattern

matching the original study (Sel et al. found the same Question × Synchrony interaction, $F(1, 28) = 4.6$, $p = .014$). A marginal main effect of Morphing was observed, $F(1, 43) = 3.46$, $p = .070$, but no significant interaction involved this factor (all interaction ps > .50). Data were therefore collapsed across Morphing for planned comparisons, as in Sel et al. (2017).

Follow-up paired t-tests per question revealed that synchronous stimulation produced higher ratings on Q2 ($t(43) = 2.33$, $p = .024$), directly replicating the key effect reported by Sel et al. (2017) (their $t(28) = 2.69$, $p = .012$ on the same item). We further extended this result by finding a significant synchrony effect on Q1 ($t(43) = 2.22$, $p = .032$), whereas Sel et al. had not observed a significant effect on this item (their $t(28) = -0.27$, $p = .79$). No synchrony effect emerged for Q3 ($t(43) = -0.70$, $p = .49$), consistent with the non-significant trend reported by Sel et al. on the same item (their $t(28) = -1.6$, $p = .12$).

Adding Lucidity as a between-subject factor to the ANOVA revealed a significant main effect of Lucidity, $F(1, 42) = 6.79$, $p = .013$, $\eta^2G = .063$, BF10 = 4.14, indicating that non-lucid dreamers reported more subjective enfacement than frequent lucid dreamers across all conditions (Q1-Q3 combined: M_nLD = 45.2, SD = 18.1; M_fLD = 31.0, SD = 18.2; $t(42) = 2.61$, $p = .013$, Hedges' $g = 0.77$), driven by Q2 ($p = .015$, $g = 0.75$) and Q3 ($p = .029$, $g = 0.67$). The Question × Synchrony interaction previously reported persisted, $F(1.62, 67.98) = 4.37$, $p = .023$, but no higher-order interaction involving Lucidity reached significance (Lucidity × Synchrony: $p = .259$; BF_inclusion = 0.44; Lucidity × Question × Synchrony: $p = .364$), indicating that both groups were comparably affected by the synchrony manipulation at the aggregate level. No group differences were observed for the pulsing items (Q4-Q5, all p > .39) or distraction (Q6, p = .**81).**

Discussion

We replicated the cardio-visual enfacement illusion, reproducing the core findings of Sel et al. (2017). Cardio-visual stimulation shifted self-other perception, with participants judging morphed faces as more self-like following stimulation. As in the original study, the specific effect of cardiac synchrony stimulation on self-face recognition emerged only when interoceptive accuracy was included as a covariate, in line with the view that the cardio-visual enfacement illusion is modulated by individuals' sensitivity to perceive their own cardiac signals.

However, the pattern that emerged in our sample diverged from the original. Sel et al. observed the strongest illusion among high-IAcc participants under cardio-synchronous stimulation with the other-like face. In our data, a dissociation emerged across the interoceptive accuracy continuum. At low IAcc, we closely replicated Sel et al. (2017): participants showed the classic illusion with the other-like stimulation face. At high IAcc, however, the pattern shifted: the original synchronous other-like effect was no longer significant, while significant shifts emerged in the synchronous self-like and asynchronous other-like conditions, suggesting a broadening and redistribution of the illusion rather than a simple reversal.

This dissociation may reflect the composition of our sample, which differed from the general population tested by Sel et al. (2017). The inclusion of frequent lucid dreamers, who, to our knowledge, were tested here for the first time on the heartbeat counting task (Schandry, 1981), shifted the IAcc distribution of our sample upward. Frequent lucid dreamers showed higher interoceptive accuracy than non-lucid dreamers, and the full-sample IAcc continuum therefore conflated two populations with distinct interoceptive profiles. Whether the observed IAcc advantage in frequent lucid dreamers reflects a genuinely superior interoceptive precision or partly depends on beliefs about heart rate, as suggested by recent critiques of the heartbeat counting task (Zamariola et al., 2018; Corneille et al., 2020), cannot be fully disentangled here and will be addressed in our ongoing heartbeat discrimination study.

Adding lucid dreaming status as a grouping factor revealed that the two populations expressed different patterns across the IAcc continuum. Non-lucid dreamers, who clustered at the lower end of the IAcc distribution, progressively extended the illusion beyond the classic synchronous other-like condition: at high IAcc, their self-other boundaries appeared broadly permeable across multiple stimulation conditions. Frequent lucid dreamers showed a contrasting pattern on the other-like face: as interoceptive accuracy increased, the illusion weakened and reversed, with participants perceiving the morphed face as less self-like after cardio-visual stimulation than before. Within a predictive processing framework, this pattern could be consistent with a more stable or more precisely weighted self-model in frequent lucid dreamers, in which prediction errors from the exteroceptive signal are weighted less heavily in the posterior update and more readily attributed to an external source than integrated into the self-representation. This reversal was however supported by only anecdotal-to-moderate Bayesian evidence (BF10 = 2.41) and calls for replication in a larger sample. Its interpretation as a trait-level signature of lucid dreaming therefore remains tentative.

The group difference between frequent and non-lucid dreamers was also reflected in subjective reports. Non-lucid dreamers rated the enfacement experience substantially higher than frequent lucid dreamers across all conditions ($F(1, 42) = 6.79$, $p = .013$, BF10 = 4.14, Hedges' $g = 0.77$). Across-participant correlations between subjective ratings and behavioral ΔPSE did not reach significance ($r = .12$, $p = .46$ in the full sample; $r = .29$, $p = .18$ in frequent lucid dreamers), consistent with the dissociation reported by Sel et al. (2017), who observed that interoceptive accuracy correlated with behavioral but not with subjective measures. Subjective questionnaires rely on introspective evaluations and constitute a coarser measure than the fine-grained ΔPSE. Nonetheless, each measure independently captured the group difference, with frequent lucid dreamers showing both reduced behavioral illusion and reduced subjective enfacement, providing convergent evidence of attenuated illusion susceptibility.

Within the frequent lucid dreaming group, LuCiD scores and interoceptive accuracy did not correlate ($r = .06$, $p = .80$), suggesting that self-reported lucidity intensity and objective interoceptive precision capture distinct facets of the lucid dreaming phenotype. Both dimensions differentiated frequent from non-lucid dreamers at the

group level but varied independently within the frequent lucid dreaming group. This is consistent with the view that lucid dreaming is a multidimensional construct, with interoceptive precision and metacognitive awareness potentially representing distinct contributing components (Simor et al., 2022).

Several limitations should be acknowledged. First, interoceptive accuracy was measured using the heartbeat counting task (Schandry, 1981), which we adopted for comparability with Sel et al. (2017). This task has been criticized for conflating interoceptive sensitivity with beliefs about heart rate (Zamariola et al., 2018; Corneille et al., 2020), and its validity as a pure index of interoceptive accuracy remains debated (Murphy et al., 2019). A follow-up study using the heartbeat discrimination task (Brener & Ring, 2016), which controls for these confounds and yields a metacognitive m-ratio (Garfinkel et al., 2015), is currently underway in our lab. Second, the exploratory model including Lucidity involved a relatively large number of parameters for the available sample size (n = 22 per group), and the Synchrony × IAcc × Lucidity interaction was attenuated under more conservative random-effects specifications (see Supplementary Materials). The group-specific findings therefore warrant replication in larger, pre-registered samples. Third, although subjective reports revealed a clear group-level difference in overall enfacement, the Lucidity × Synchrony interaction did not reach significance in the questionnaire ANOVA ($p = .259$, $BF_{inclusion} = 0.44$), and any apparent higher subjective sensitivity to synchrony in frequent lucid dreamers should therefore not be over-interpreted. Fourth, because the inclusion of frequent lucid dreamers raised the sample mean, "high IAcc" in our centered analyses (+1.5 SD) corresponds to absolute IAcc values above those typically represented in Sel et al.'s (2017) sample, and part of the observed divergence may therefore reflect a broader coverage of the IAcc continuum rather than a fundamentally different pattern at matched absolute IAcc. Relatedly, a sensitivity analysis with IAcc recentered within the non-lucid dreamers subgroup showed that the 3-way Synchrony × Morphing × IAcc interaction was attenuated ($F(1, 60) = 2.11$, $p = .15$) when between-group variance in IAcc was removed. This is expected given the reduced range and lower power within this subgroup (n = 22) and is consistent with the hypothesis that the IAcc-driven modulation reflects trait differences between populations rather than within-group variance. It does not affect the primary full-sample analysis ($F(1, 126) = 8.56$, $p = .004$), which preserves the between-group variance of theoretical interest (Enders & Tofighi, 2007). Finally, the present findings are limited to a single multisensory paradigm. Two experiments currently underway in our lab extend this work to the rubber-hand illusion combined with heartbeat discrimination, and to a rod-and-frame paradigm in virtual reality, which will test whether interoception-driven resistance to self-model updating generalizes across multisensory illusions.

Taken together, these results suggest that interoceptive precision shapes how individuals resolve multisensory conflicts about the self, and that frequent lucid dreamers differ from non-lucid dreamers in the direction and magnitude of this modulation. Future studies combining more refined metacognitive indices of interoception with other multisensory paradigms, and formal computational modeling

of perceived self-similarity under multisensory conflict, will help clarify whether the waking differences observed here relate to the predictive mechanisms proposed to support lucid dreaming (Simor et al., 2022).

References

Bucci, A., & Grasso, M. (2017). Sleep and dreaming in the predictive processing framework.

Filevich, E., Dresler, M., Brick, T.R., Kühn, S. (2015). Metacognitive mechanisms underlying lucid dreaming. J. Neurosci. 35, 1082–1088.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature reviews neuroscience*, *11*(2), 127-138.

Hohwy, J. (2013). The predictive mind. OUP Oxford.

A. Clark, Surfing Uncertainty: Prediction, Action, and the Embodied Mind (Oxford University Press, 2016).

Grush, R. (2004). The emulation theory of representation: Motor control, imagery, and perception. *Behavioral and brain sciences*, *27*(3), 377-396.

A. Clark, Whatever next? Predictive brains, situated agents, and the future of cognitive science. Behav. Brain Sci. 36, 181–204 (2013).

Windt, J. M. (2015). *Dreaming: A conceptual framework for philosophy of mind and empirical research*. MIT press.

Llewellyn, S. (2016). Dream to predict? REM dreaming as prospective coding. *Frontiers in Psychology*, *6*, 1961.

Hobson, J. A., & Friston, K. J. (2012). Waking and dreaming consciousness: neurobiological and functional considerations. Progress in neurobiology, 98(1), 82-98.

Hobson, J. A., Pace-Schott, E. F., & Stickgold, R. (2000). Dreaming and the brain: toward a cognitive neuroscience of conscious states. *Behavioral and brain sciences*, *23*(6), 793-842.

Nir, Y., & Tononi, G. (2010). Dreaming and the brain: from phenomenology to neurophysiology. *Trends in cognitive sciences*, *14*(2), 88-100.

Van Eeden, F. (1913, July). A study of dreams. In *Proceedings of the society for psychical research* (Vol. 26, No. 47, pp. 431-461).

La Berge, S.P., Nagel, L.E., Dement, W.C., and Zarcone, V.P. (1981). Lucid Dreaming Verified by Volitional Communication during Rem Sleep. Percept. Mot. Skills 52, 727–732.10.2466/pms.1981.52.3.727.

La Berge, S. P. (1980). Lucid dreaming as a learnable skill: A case study. *Perceptual and motor skills*, *51*(3_suppl2), 1039-1042.

Simor, P., Bogdány, T., & Peigneux, P. (2022). Predictive coding, multisensory integration, and attentional control: A multicomponent framework for lucid dreaming. Proceedings of the National Academy of Sciences, 119(44), e2123418119.

Talsma, D. (2015). Predictive coding and multisensory integration: an attentional account of the multisensory mind. *Frontiers in integrative neuroscience*, *9*, 19.

Gregory, R. L. (1997). Knowledge in perception and illusion. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, *352*(1358), 1121-1127.

Sel, A., Azevedo, R. T., & Tsakiris, M. (2017). Heartfelt self: cardio-visual integration affects self-face recognition and interoceptive cortical processing. Cerebral Cortex, 27(11), 5144-5155.

Baird, B., Castelnovo, A., Gosseries, O., & Tononi, G. (2018). Frequent lucid dreaming associated with increased functional connectivity between frontopolar cortex and temporoparietal association areas. Scientific reports, 8(1), 17798.

B. Baird, S. A. Mota-Rolim, M. Dresler, The cognitive neuroscience of lucid dreaming. Neurosci. Biobehav. Rev. 100, 305–323 (2019).

Simons, J. S., Garrison, J. R., & Johnson, M. K. (2017). Brain Mechanisms of Reality Monitoring. *Trends in Cognitive Sciences*, *21*(6), 462–473. https://doi.org/10.1016/j.tics.2017.03.012

Muzur, E. F. Pace-Schott, J. A. Hobson, The prefrontal cortex in sleep. Trends Cogn. Sci. 6, 475–481 (2002).

Maquet, P., Ruby, P., Maudoux, A., Albouy, G., Sterpenich, V., Dang-Vu, T., Desseilles, M., Boly, M., Perrin, F., Peigneux, P., & Laureys, S. (2005). Human cognition during REM sleep and the activity profile within frontal and parietal cortices: A reappraisal of functional neuroimaging data. In *Progress in Brain Research* (Vol. 150, pp. 219–595). Elsevier. https://doi.org/10.1016/S0079-6123(05)50016-5

Szymanek, Piotr. 2021. "I Did Not Expect to Be Dreaming: Explaining Realization in Lucid Dreams with a Bayesian Framework." *Consciousness and Cognition* 93: 103163. doi:10.1016/j.concog.2021.103163.

LaBerge, S., & Rheingold, H. (1991). *Exploring the world of lucid dreaming*. Ballantine Books.

Check: Kanani & Padole (2018), Brener et al. (1993), Suzuki et al. (2013)